ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL MATHEMATICS

PAPER 1

MARKS: 100 **TIME: 2 hours**

Topics: numbers and binary numbers, exponents and surds, algebra, equations, functions and graphs, finance.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL calculations clearly.
2. You may use an approved scientific calculator, unless stated otherwise. Round off to TWO decimal places where necessary.

QUESTION 1: NUMBERS

1.1 Classify each number as rational, irrational or non-real: $\sqrt{9}$; π ; $-¾$; $\sqrt{-4}$ (4)

1.2 Convert 45 (base 10) to a binary number. (2)

1.3 Convert 11010_2 to a decimal number. (2)

1.4 Calculate $1011_2 + 110_2$. Give your answer as a binary number. (2)

1.5 Round off 3,14159 to THREE decimal places. (1)

1.6 Write 0,000 045 in scientific notation. (2)

1.7 Between which two consecutive integers does $\sqrt{50}$ lie? (2)

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QUESTION 2: EXPONENTS, SURDS AND ALGEBRA

2.1 Simplify:

2.1.1 $(2a^3b^2)^3$ (2)

2.1.2 $\dfrac{x^5 \cdot x^{-2}}{x^2}$ (3)

2.1.3 $\sqrt{(16x^8)}$ (2)

2.1.4 $2\sqrt{12} + \sqrt{27}$ (3)

2.2 Factorise fully:

2.2.1 $x^2 - 7x + 12$ (2)

2.2.2 $4x^2 - 25$ (2)

2.2.3 $3x^2 + 6x$ (2)

2.3 Simplify $\dfrac{x^2 - 4}{x + 2}$ (3)

2.4 Expand and simplify $(x + 4)(2x - 3)$ (3)

2.5 Simplify $3(2x - 1) - 2(x + 4)$ (3)

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QUESTION 3: EQUATIONS AND INEQUALITIES

3.1 Solve for x:

3.1.1 $5x - 3 = 2x + 9$ (2)

3.1.2 $x^2 - x - 6 = 0$ (3)

3.1.3 $2^x = 32$ (2)

3.2 Solve and represent the solution on a number line: $3 - 2x > 7$ (3)

3.3 Solve simultaneously: $y = 2x + 1$ and $3x + y = 11$ (4)

3.4 Ohm's law is $V = IR$. Make R the subject and calculate R when V = 12 V and I = 0,5 A. (3)

3.5 The perimeter of a rectangle is $P = 2(l + b)$. Calculate b if P = 30 cm and l = 9 cm. (3)

3.6 The length of a rectangular metal plate is 3 cm more than its width. Its area is 40 cm². Calculate the width and the length of the plate. (5)

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QUESTION 4: FUNCTIONS AND GRAPHS

Given: $f(x) = -x^2 + 9$ and $g(x) = x + 3$

4.1 Write down the coordinates of the y-intercept of f. (1)

4.2 Calculate the coordinates of the x-intercepts of f. (2)

4.3 Calculate the coordinates of the x-intercept of g. (1)

4.4 Write down the equation of the axis of symmetry of f, and the gradient of g. (2)

4.5 Sketch the graphs of f and g on the same set of axes. Show all intercepts and the turning point. (5)

4.6 Calculate the coordinates of the points where f and g intersect. (4)

4.7 Write down the range of f. (1)

4.8 For which values of x is $f(x) \geq g(x)$? (2)

4.9 For $h(x) = \frac{4}{x}$, write down the equations of the asymptotes and the quadrants in which the graph lies. (3)

4.10 For $k(x) = 3^x$, write down the y-intercept and the equation of the asymptote. (2)

4.11 Write down the equation of a straight line parallel to g that passes through the point (0 ; −2). (2)

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QUESTION 5: FINANCE

5.1 R15 000 is invested at 9% per annum for 4 years. Calculate the final amount if the interest is: (a) compound interest, compounded annually; (b) simple interest. (7)

5.2 Calculate the difference between the two amounts in QUESTION 5.1. (1)

5.3 A machine bought for R40 000 depreciates by 10% per year using the straight-line method. Calculate its value after 3 years. (2)

[10]

TOTAL: 100

END OF QUESTION PAPER